



An Empirical Meta-Evaluation of Language Model Evaluation Methods for Systems Engineering

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Biographies



Ryan Bell

is a PhD Candidate in Systems Engineering and a 10 year experienced engineer in the defense industry and scheduled to graduate in June.

In his current role as the Data Science and Artificial Intelligence Lead Systems Engineer at Naval Information Warfare Center Atlantic, Ryan provides modeling and simulation expertise to a variety of programs for the Navy and USMC.

He specializes in simulating communication systems in complex environments and is an advocate for the use of digital engineering early in the systems engineering lifecycle.

Ryan earned a BS in Electrical Engineering from Clemson University, a MS in Electrical Engineering from Clemson University with a focus on Electronics.

He is a South Carolina registered Professional Engineer (PE), published author, and teacher.



Raymond Madachy, PhD

is a Professor in the Systems Engineering Department at the Naval Postgraduate School.

His research interests include digital engineering tool environments, modeling and simulation of systems and software engineering processes, generative AI, system cost modeling, affordability and tradespace analysis. His research has been funded by DoD agencies, the National Security Agency, NASA, and industry. He has developed widely used tools for systems and software cost estimation including the online [COSYSMO](#) and [COCOMO](#) tools, and is founding developer of the open-source [Systems Engineering Library \(se-lib\)](#).

His books include *Software Process Dynamics*, *What Every Engineer Should Know about Modeling and Simulation*, and *What Every Engineer Should Know about Python*. He is co-author of *Software Cost Estimation with COCOMO II* and *Software Cost Estimation Metrics Manual for Defense Systems*.

He received the Jeff Grady Award for "Advancing Systems Engineering Knowledge" from the INCOSE San Diego Chapter in 2025, and the USC Center for Systems and Software Engineering Lifetime Achievement Award for "Innovative Development of a Wide Variety of Cost, Schedule and Quality Models and Simulations" in 2016.

Agenda



1. Why Language Models are Compelling for Systems Engineering
2. Language Model Benchmarks
3. SysEngBench
4. Create Multiple Choice Distractor Questions and Open Style Questions
5. Evaluation Modality Comparison Results
6. Contributions and Conclusion

Why Language Models are Compelling for Systems Engineering



MBSE helps structure engineering complexity. AI can help engineers navigate and reason across that structured information.

Has the potential to:

- Enable the transformation from **document-centric to model-centric engineering** baselines
- Support **drafting and querying** of linked SE artifacts
- Improve **traceability and cross-artifact coherence** across the digital thread



But Adoption Requires Trust



... And trust is built with evidence.

- SE needs *qualification evidence* that LLM outputs are reliable under **SE-relevant response constraints, failure modes, and consequence severity**, but the evidence practitioners typically have is **misaligned** with SE work products (Raz, 2023).
- **MCQ benchmarks** are commonly used to evaluate model readiness, even though many SE tasks require **constructive artifacts** (explanations, justifications, trade-space arguments, traceable rationale) **rather than recognition-style selection** (Tianle Li, 2024; Jin Li, 2024; Ünal, 2023).
- LLM failure modes such as hallucination, contamination, and limited transparency become especially problematic when outputs inform architecture, requirements allocation, or risk assessments (Sanu, 2024; Matarazzo, 2025; Johnson, 2025).

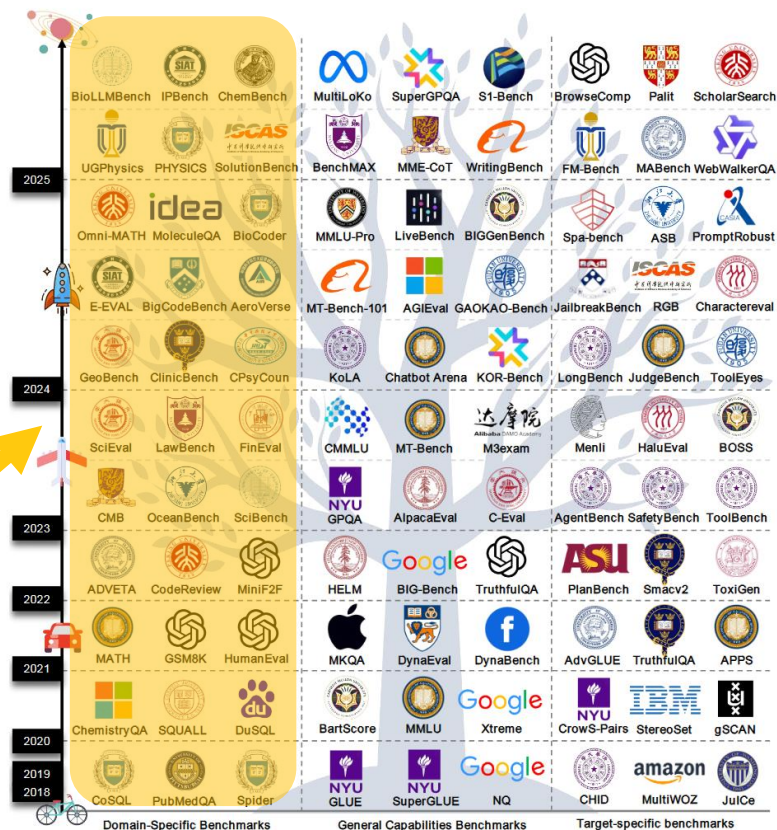
How have other domains tackled this problem?



Benchmarks!

- Benchmarks have evolved over time
- Early benchmarks focused on foundational tasks such as work relationships and their semantic similarities
- Complexity and domain specificity of the benchmarks over time have been indicative of progression

Notice Systems Engineering is missing!



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SysEngBench Examples



- Example of a successful conversion: Definition recall
- Example of Option-dependent failure mode: “Which of the following...”
- Example of No canonical answer failure mode: multiple answers could be correct.

Q6. Failed MCQ→OSQ conversion: option-dependent question

Field	Content
Original MCQ	In the context of systems engineering, which of the following best categorizes the different types of risks and system lifecycle phases? A. Lifecycle phases: Design, Manufacturing, Operation; Types of risk: Technical, Schedule, Cost. B. Lifecycle phases: Planning, Testing, Maintenance; Types of risk: Environmental, Legal, Market. C. Lifecycle phases: Research, Development, Deployment; Types of risk: Operational, Financial, Strategic. D. Lifecycle phases: Conceptualization, Execution, Closure; Types of risk: Human, Physical, Informational. Answer: A
Suitability Score	2 out of 10 (below confidence threshold)
Rejection Rationale	The question depends on the provided options to specify which taxonomy (e.g., INCOSE, NASA/DoD) is intended, so without choices there is no single canonical answer and major ambiguity is introduced. Removing the options alters the construct from selecting the best categorization to recalling an unspecified framework, compromising intent and semantics despite a potentially short response length.

Q13. Failed MCQ→OSQ conversion: recognition-to-generation drift

Field	Content
Original MCQ	Which of the following is an example of a “Condition–Consequence” risk statement in systems engineering? A. IF the project exceeds its budget by 20%, THEN additional funding will be required. B. A strong project management team ensures project success. C. High-quality components decrease the likelihood of system failure. D. Effective communication is key to project management. Answer: A
Suitability Score	4 out of 10 (below confidence threshold)
Rejection Rationale	Without options, there is no single canonical answer—many examples could fit—so validity hinges on subjective pattern matching. Removing the options shifts the construct from recognition to generation, introducing ambiguity, though the expected response could be brief. This creates a risk of construct drift despite a tight scope.

Field	Content
Original MCQ	What is the primary purpose of stakeholder analysis in systems engineering? A. To identify project funding sources B. To understand stakeholder needs and concerns C. To assign project responsibilities D. To develop marketing strategies Answer: B
OSQ Prompt	What is the primary purpose of stakeholder analysis in systems engineering?
Expected Answer	To understand stakeholder needs and concerns.
Bloom’s Level	Remember (definitional recall).

Note: This example focuses on metadata that was used from the MCQ → OSQ conversion. Full/partial/no-credit rubric criteria and Bloom’s justification were collected as metadata for future use, but are omitted here for brevity.

So how did the models do?



- SysEngBench was ran on 35 LLMs, benchmarked, and plotted on a Pareto curve
- Efficiency spotted with mid-sized models with Phi-3.5-mini-instruct and similar 3-4B parameter models deliver ~88 % accuracy, offering strong performance at a fraction of the compute budget.

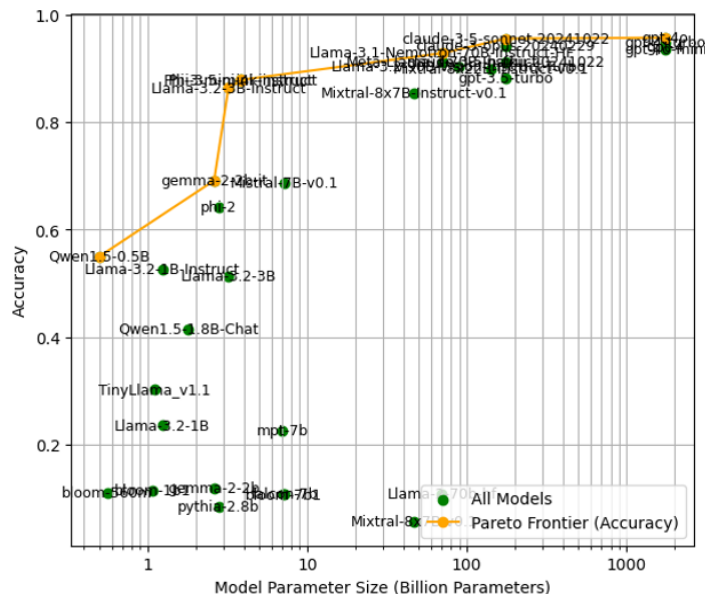


TABLE 4 Performance on SysEngBench by Category

LLM	Score ¹	SEO ²	Lp ³	TM ⁴	TMp ⁵	CCSEB ⁶	SEA ⁷
claude-3-5-sonnet-20241022	0.96	0.95	0.93	0.96	1.00	0.97	0.94
gpt-4o	0.96	0.95	0.95	0.96	0.97	0.96	0.97
gpt-4o-turbo	0.95	0.93	0.92	0.95	1.00	0.97	0.95
gpt-4	0.94	0.92	0.92	0.95	0.97	0.96	0.97
gpt-4o-mini	0.94	0.91	0.93	0.94	0.99	0.96	0.98
claude-3-opus-20240229	0.94	0.92	0.92	0.95	0.99	0.95	0.97
Llama-3.1-Nemotron-70B-Instruct-HF	0.93	0.91	0.87	0.95	0.98	0.95	0.94
Llama-3.2-90B-Vision-Instruct-Turbo	0.91	0.89	0.86	0.92	0.99	0.89	0.92
Meta-Llama-3-70B-Instruct	0.91	0.90	0.87	0.93	0.98	0.88	0.96
claude-3-5-haiku-20241022	0.91	0.87	0.89	0.92	0.99	0.95	0.92
Meta-Llama-3.1-405B-Instruct-Turbo	0.90	0.91	0.84	0.91	0.91	0.88	0.89
Mixtral-8x22B-Instruct-v0.1	0.90	0.86	0.86	0.91	0.99	0.91	0.96
Phi-3-mini-4k-instruct	0.88	0.82	0.86	0.89	0.95	0.91	0.93
gpt-3.5-turbo	0.88	0.82	0.86	0.89	0.96	0.88	0.96
Phi-3.5-mini-instruct	0.88	0.81	0.85	0.89	0.97	0.92	0.93
Llama-3.2-3B-Instruct	0.87	0.82	0.82	0.88	0.96	0.85	0.95
Mixtral-8x7B-Instruct-v0.1	0.86	0.79	0.83	0.86	0.95	0.86	0.92
gemma-2-2b-it	0.69	0.59	0.76	0.68	0.83	0.71	0.73
Mistral-7B-v0.1	0.68	0.58	0.75	0.69	0.74	0.79	0.64
phi-2	0.64	0.54	0.81	0.61	0.66	0.67	0.69

ORIGINAL ARTICLE
Journal Section

Baselining Large Language Model Performance in Systems Engineering Using SysEngBench

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In the rapidly evolving field of Artificial Intelligence (AI), Large Language Models (LLMs) have demonstrated impressive capabilities in generating natural language. However, their proficiency in specialized domains, particularly in the field of systems engineering, remains less explored and unquantified. This paper introduces SysEngBench, a novel benchmark specifically designed to evaluate LLMs in the context of systems engineering concepts and applications. SysEngBench encompasses a comprehensive set of tasks derived from core systems engineering processes, including requirements analysis, system architecture design, risk management, and stakeholder communications, to provide an assessment of language model abilities.

Our evaluation of leading LLMs using SysEngBench reveals a strong correlation between model scale and performance, with state-of-the-art models like GPT-4o and Claude 3.5 Sonnet achieving near-human-level accuracy across multiple categories. Smaller models like Llama 3.2 3B exhibit significantly higher defect densities and lower accuracy scores, highlighting their limitations in handling systems engineering tasks. However, Pareto analysis demonstrates that while larger models generally outperform smaller ones, some mid-sized models like Phi-3.5-mini-instruct achieve competitive accuracy with significantly lower computational requirements. These findings suggest pathways for practitioners and future research, particularly in optimizing smaller models for domain-specific reasoning through fine-tuning or knowledge integration.

SysEngBench provides a systematic approach to assessing AI's impact on systems engineering, offering insights into the trade-offs between model efficiency and accuracy. By establishing a benchmark for LLM evaluation in this domain, we provide a cohesive, extensible, and effective method to refine AI's role in the systems engineering discipline.

KEYWORDS

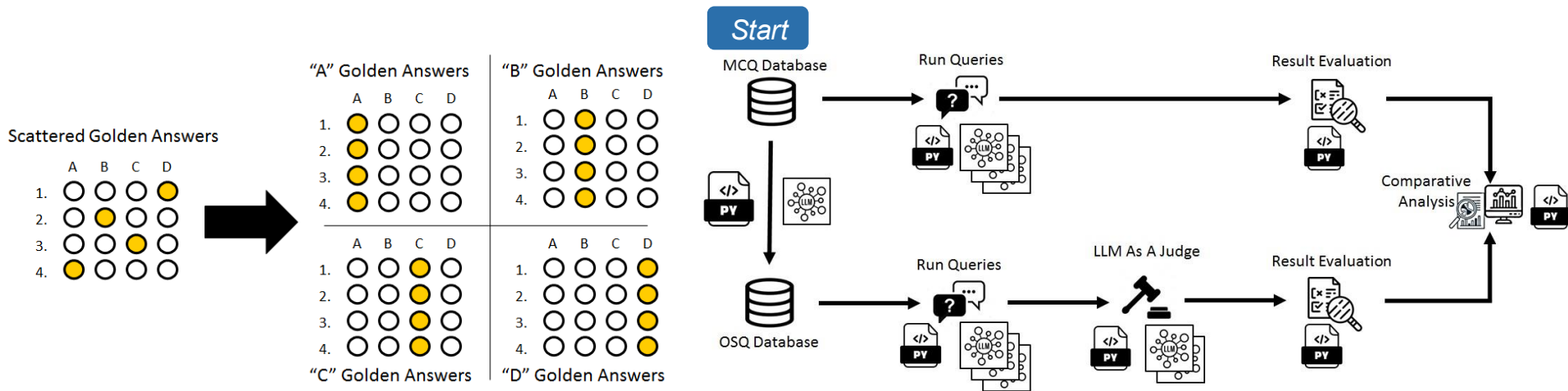
Systems Engineering, Large Language Models, Benchmarking, SysEngBench, Performance Evaluation

R. Bell, R. Longshore, R. Madachy, and A. Hanrahan, "Baselining large language model performance in systems engineering using SysEngBench," *Systems Engineering* Wiley, 2026, doi: [10.1002/sys.70067](https://doi.org/10.1002/sys.70067).

Is a Multiple Choice Question (MCQ) test the best method?



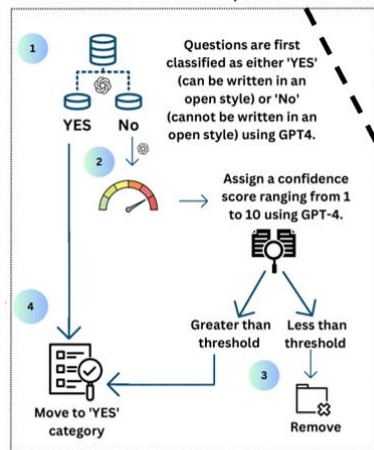
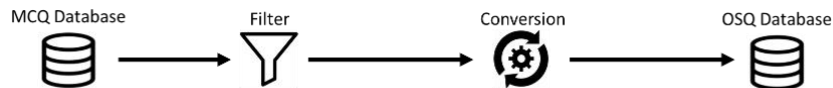
- What if we changed the position of the correct answer for the MCQs?
 - This is straight-forward, we simply shift the golden answers.
- What if we made Open Style Questions (OSQs) that asked the same question as their MCQ counter-parts, just without viable answers?
 - We'll need to convert questions at scale. Language models can help us here.



Converting MCQ to OSQ Approach and Conversion Results



- Use filtering and conversion prompts with a ChatGPT
- The accepted region contained roughly three-quarters of the questions



Conversion Prompt

Convert the following multiple-choice question into a textual question while fully preserving the original in semantic clarity. Refactor the question so that it is a textual response without requiring answer choices or correctness. Ensure the rephrased question tests the same knowledge as the original multiple-choice question.

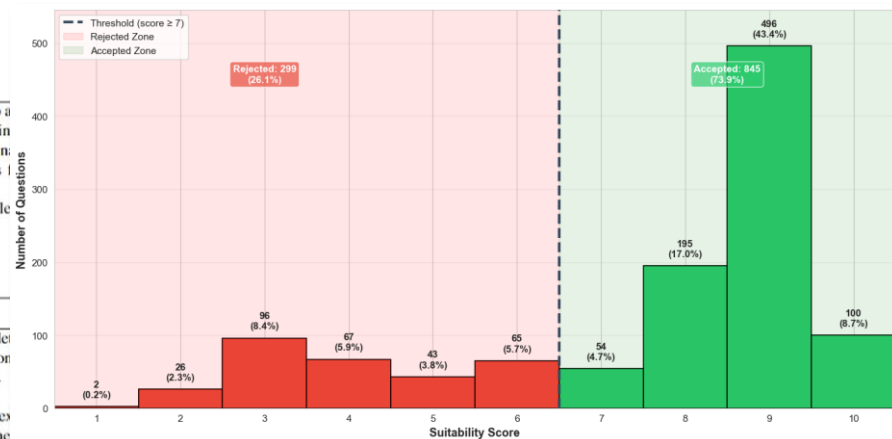
Question: {question}
Reference Answer: {answer}
Converted Open-Ended Question: {response}

Filter Prompt

Carefully evaluate the following multiple-choice question and determine whether it can be accurately converted into an open style question without losing its original intent, meaning, or semantic clarity. Specifically, ensure that:

- The core concept being tested can be preserved in free-text.
- The question does not rely on answer choices to provide context or disambiguation.
- The open-ended format would still allow for a clear, concise, and unambiguous response.

Answer 'Yes' if the question is suitable for conversion while maintaining intent and semantics. Otherwise, answer 'No'. In both cases, the 'Yes' or 'No' response should be followed by a confidence score between 1-10 for your confidence on whether the question can be converted or not.



OSQ Scoring with LLM-as-a-Judge



- A multi-dimension rubric design was adopted as it:
 - Scores each OSQ response against a canonical reference answer.
 - Converts open-ended responses into structured 0–100 quality scores.
 - Separates dimensions for more detailed analysis.
 - Scales linearly with dataset size.

LLM as a Judge Scoring Rubric

Dimension	Description	Points
Technical Accuracy	Correctness of facts, principles, terminology	20
Conceptual Understanding	Depth of systems thinking demonstrated	20
Completeness	Coverage of required answer elements	20
Clarity	Communication quality and organization	20
Professional Relevance	Relevance to the intended SE concept or context	20

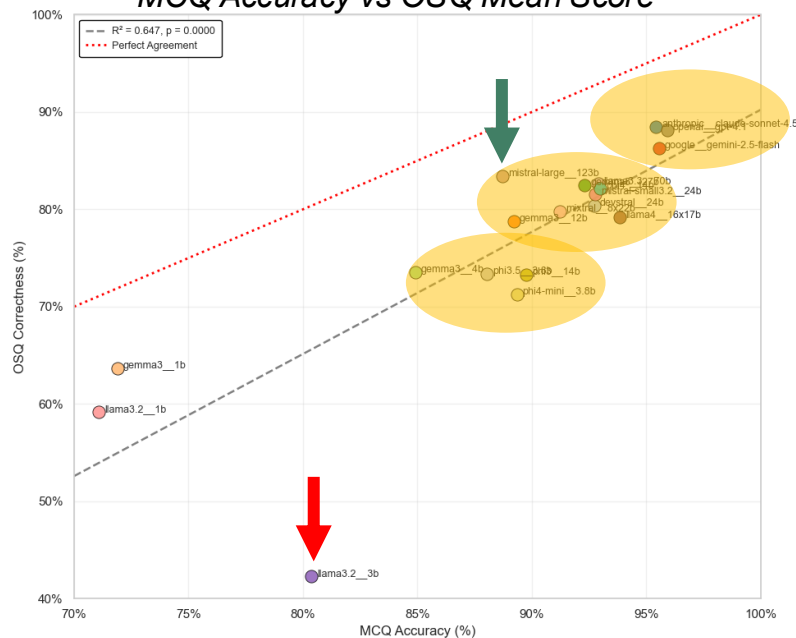
- Dimension selection matches similar evaluation dimensions in literature:
 - G-Eval (Liu et al. 2023): coherence, consistency, fluency, relevance
 - LLM-Eval (Lin & Chen 2023): appropriateness, content, grammar, relevance
 - MT-Bench (Zheng et al. 2023): accuracy, depth, level of detail, helpfulness
 - Prometheus 2 (Kim et al. 2024): fine-grained user-supplied rubrics with per-criterion scoring

Comparing MCQ and OSQ Modalities



- Clusters forming at different centroids around 87%, 92%, and 97%
- Clear trend of lower scores on OSQ when compared to MCQ
- Deltas ranged from lowest of 5% to highest of 38%

MCQ Accuracy vs OSQ Mean Score



Per-Model MCQ–OSQ Performance Differences, Sorted by Δ_i (Descending)

Model	MCQ Accuracy	OSQ Mean (norm.)	Δ_i
llama3.2__3b	0.804	0.423	0.381
phi4-mini__3.8b	0.893	0.713	0.181
phi3__14b	0.898	0.733	0.165
phi3.5__3.8b	0.880	0.733	0.147
llama4__16x17b	0.938	0.792	0.146
devstral__24b	0.927	0.804	0.123
llama3.2__1b	0.711	0.592	0.119
mixtral__8x22b	0.912	0.798	0.115
gemma3__4b	0.849	0.735	0.114
mistral-small3.2__24b	0.928	0.815	0.113
phi4__14b	0.930	0.822	0.108
gemma3__12b	0.892	0.787	0.105
llama3.3__70b	0.929	0.826	0.103
gemma3__27b	0.923	0.824	0.099
google_gemini-2.5-flash	0.956	0.863	0.093
gemma3__1b	0.719	0.637	0.082
openai_gpt-4.1	0.959	0.881	0.078
anthropic_claude-sonnet-4.5	0.954	0.885	0.070
mistral-large__123b	0.887	0.834	0.053

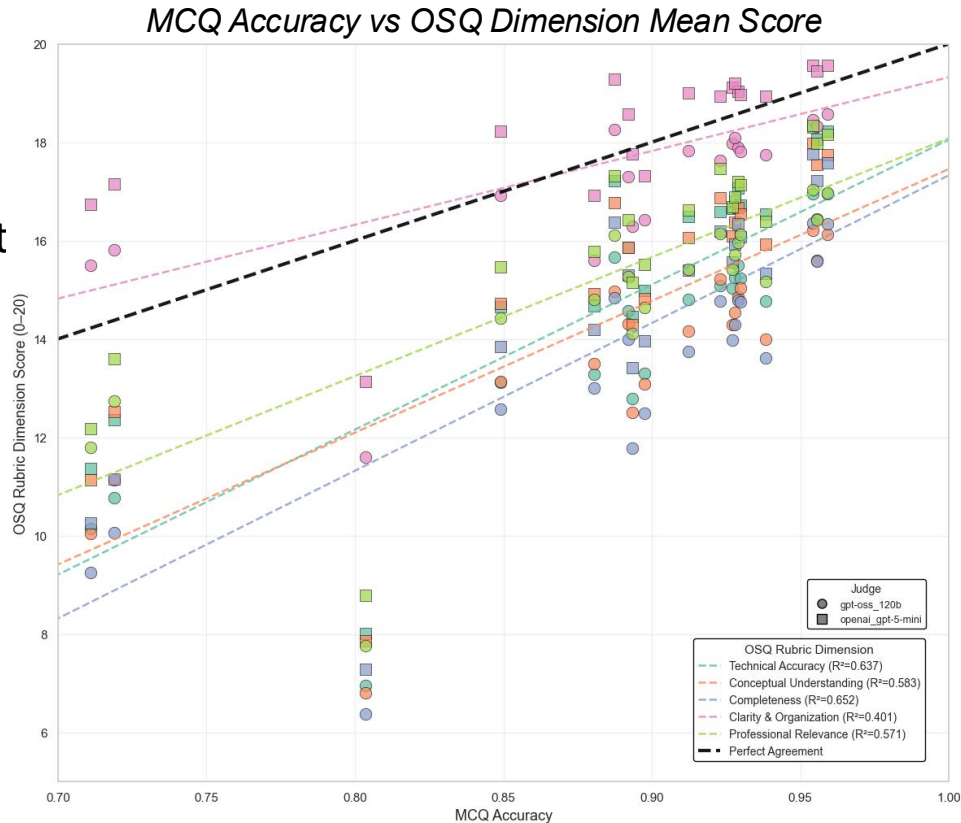
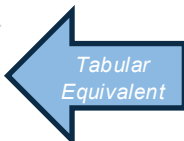
Comparing MCQ and OSQ Modalities With Rubric Dimensions



- OSQ scores were decomposed into rubric-level dimensions across both judges.
- Clarity and Organization** is the highest scoring dimension, indicating that most model outputs are readable and well-structured, even when the answer is incomplete or technically weak.
- OSQ evaluation must **separate fluency from engineering correctness**.

*MCQ vs. OSQ Rubric-Dimension
Correlation Statistics (Multi-Judge)*

Dimension	Pearson r	R^2	Slope	Intercept
Technical Accuracy	0.7979	0.6367	29.4764	-11.4247
Conceptual Understanding	0.7632	0.5825	26.7942	-9.3413
Completeness	0.8072	0.6516	30.0302	-12.7040
Clarity & Organization	0.6330	0.4006	15.0022	4.3181
Professional Relevance	0.7558	0.5712	24.1885	-6.1067



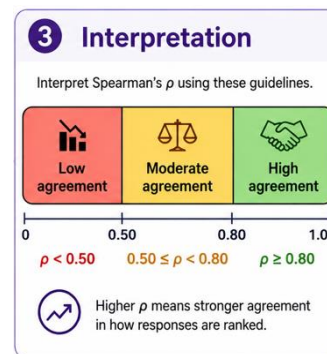
Comparing MCQ and OSQ Modalities With Model Ranking



- MCQ and OSQ agree on model ranking.
 - Across 19 models, MCQ accuracy and mean OSQ score strongly correlate and hold independently for both judges.
 - Judge choice affects the **magnitude** of the MCQ–OSQ gap but not its **existence** or the rank-order signal.
- But they disagree on absolute scores.
 - Paired Wilcoxon signed-rank tests reject equality of MCQ and OSQ scores on the same 845 matched items ($p \approx 4 \times 10^{-6}$), with every model shifting in the same direction: MCQ > OSQ.
 - Median gap: 0.08–0.14 score points depending on judge.
- Takeaway: Correlation $\neq$ interchangeability.
 - MCQ is a defensible proxy for *which* model is stronger overall, but it systematically overstates absolute performance relative to generative evaluation. This motivates dual-modality reporting.

MCQ vs. OSQ Judge Correlation Across Judges

Judge	Pearson r	Pearson p	Spearman ρ	Spearman p	n
gpt-oss_120b	0.797552	0.000043	0.800000	0.000039	19
openai_gpt-5-mini	0.810574	0.000026	0.814035	0.000022	19
Average	0.804475	0.000033	0.807018	0.000030	19



Wilcoxon Signed-Rank Test Results

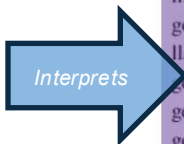
Judge	p -value	Median Difference	Mean Difference	n
gpt-oss_120b	0.000004	0.143763	0.160212	19
openai_gpt-5-mini	0.000004	0.076000	0.091734	19
Average	0.000004	0.112503	0.125970	19

Comparing MCQ and OSQ Modalities With Statistics



- **No model treats MCQ and OSQ as equivalent.**
 - Every model lands in Divergent (6) or Mixed evidence (13)
- **Gap scales inversely with capability.**
 - Smaller / older models more likely to be Divergent
- **Frontier models reduce the gap, but do not close it.**

Classification	Association (Pearson r , Spearman ρ_s)	Wilcoxon (p -value)	Effect Size ($ d $)	Takeaway / Action
INTERCHANGEABLE	High $r \geq 0.80$ and $\rho_s \geq 0.80$	Not significant $p \geq 0.05$	Small $ d < 0.50$	Similar evidence. Compare or combine.
CORRELATED BUT NON-INTERCHANGEABLE	High $r \geq 0.80$ and $\rho_s \geq 0.80$	Significant $p < 0.05$	Medium-Large $ d \geq 0.50$	Same ranking, different score levels. Report separately.
PARTIALLY ALIGNED	Moderate Both ≥ 0.50 , at least one < 0.80	Any (either)	Small-Medium $ d < 0.80$	Partial overlap. Compare cautiously.
DIVERGENT	Low $r < 0.50$ or $\rho_s < 0.50$	Significant $p < 0.05$	Medium-Large $ d \geq 0.50$	Different signals, confirmed offset. Do not combine.
MIXED / INCONCLUSIVE	Any	Any	Any	Unclear evidence. Report as ambiguous.



MCQ vs. OSQ Cross-Modality Comparison
Results by Model Grouped by Classification

Model	Pearson r	Spearman ρ_s	$ d $	p_{Wilcoxon}	Classification
Classification = Divergent					
llama3.2__3b	0.05	0.01	0.83	< 0.001	Divergent
phi4-mini__3.8b	0.41	0.37	0.68	< 0.001	Divergent
phi3__14b	0.31	0.23	0.58	< 0.001	Divergent
llama4__16x17b	0.28	0.15	0.57	< 0.001	Divergent
devstral__24b	0.41	0.26	0.53	< 0.001	Divergent
phi3.5__3.8b	0.40	0.33	0.52	< 0.001	Divergent

Classification = Mixed evidence (low association + small d)

phi4__14b	0.44	0.30	0.49	< 0.001	Mixed evidence
mistral-small3.2__24b	0.34	0.17	0.46	< 0.001	Mixed evidence
mistral__8x22b	0.35	0.27	0.45	< 0.001	Mixed evidence
google__gemini-2.5-flash	0.33	0.14	0.45	< 0.001	Mixed evidence
llama3.3__70b	0.33	0.17	0.41	< 0.001	Mixed evidence
gemma3__27b	0.38	0.31	0.41	< 0.001	Mixed evidence
gemma3__4b	0.44	0.40	0.38	< 0.001	Mixed evidence
gemma3__12b	0.41	0.32	0.38	< 0.001	Mixed evidence
openai__gpt-4.1	0.28	0.13	0.37	< 0.001	Mixed evidence
llama3.2__1b	0.41	0.41	0.36	< 0.001	Mixed evidence
anthropic__claude-sonnet-4.5	0.33	0.12	0.35	< 0.001	Mixed evidence
gemma3__1b	0.45	0.44	0.26	< 0.001	Mixed evidence
mistral-large__123b	0.41	0.24	0.21	< 0.001	Mixed evidence

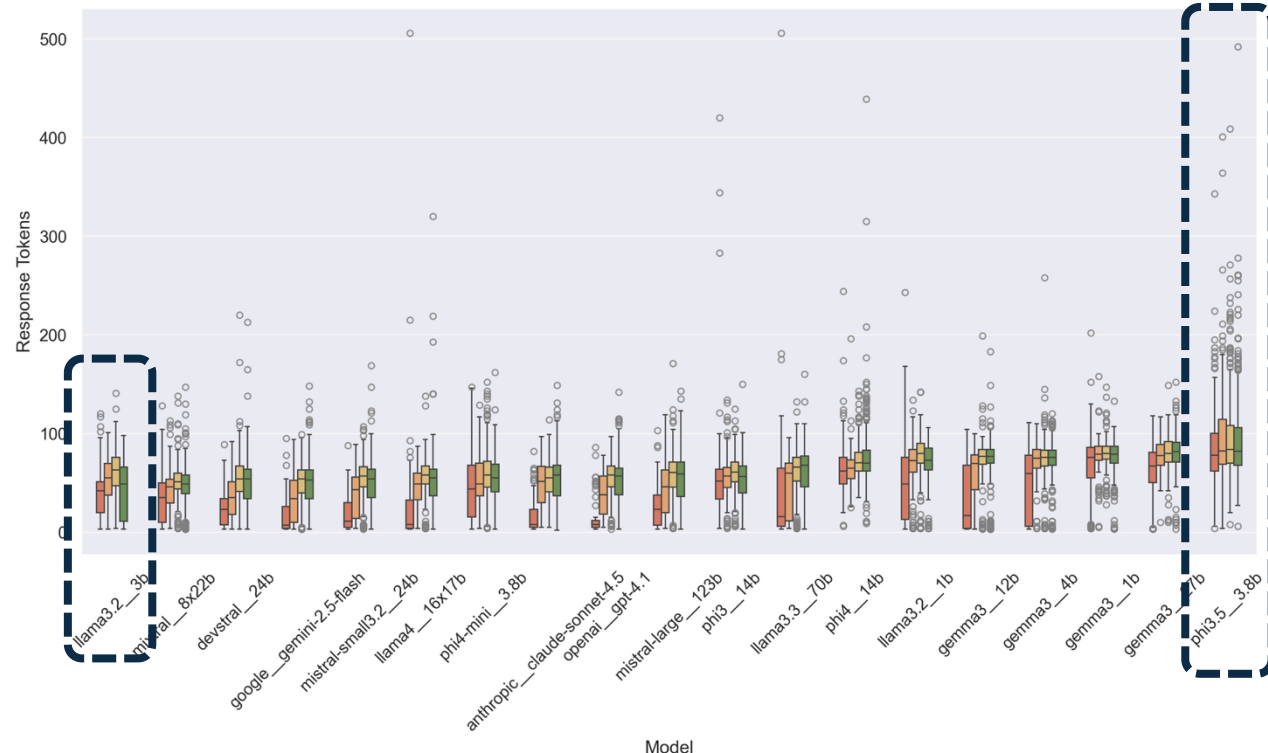
Note. r = Pearson correlation; ρ_s = Spearman rank correlation; $|d|$ = paired standardized effect-size magnitude (from paired Cohens d); p_{Wilcoxon} = two-sided paired Wilcoxon signed-rank test p -value.

OSQ Performance Versus Token Efficiency



Token Count Distributions by Model and Quality Categories

Quality
Poor (<50) Fair (50–70) Good (70–90) Excellent (90–100)



- Length of the responses measured in **tokens**
- Response quality** was broken down into Poor, Fair, Good, and Excellent ranges.
- Distribution varies substantially across models
 - Even across similar size models

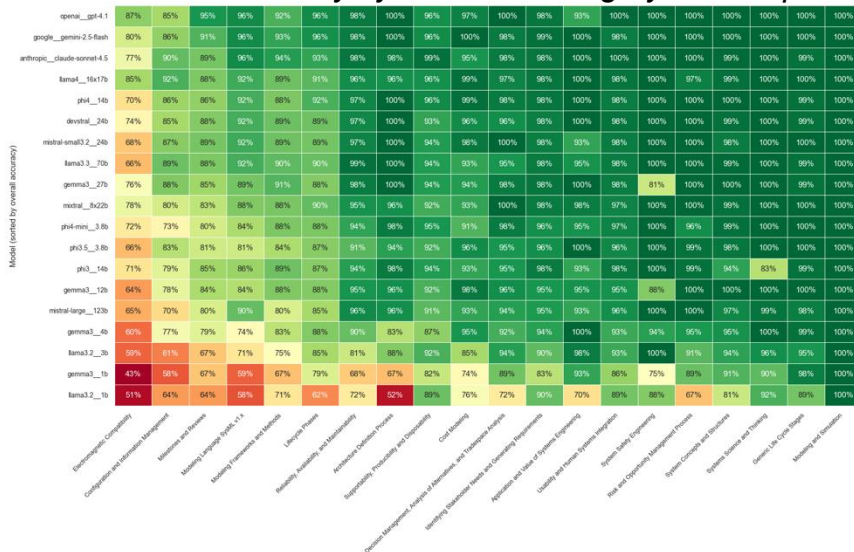
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- Pareto Efficiency Frontier: Quality vs. Token Usage**
- The plot displays the performance of various LLM models based on their Mean Judge Score (X-axis) and Median Total Tokens (Y-axis). The Efficient Pareto Frontier is highlighted by a dashed red line connecting the Pareto-optimal models.
- Legend:**
- Dominated Models
 - Pareto-Optimal Models
 - - - Efficient Pareto Frontier
- Approximate Data Points from the Plot:**
- | Model | Mean Judge Score (X) | Median Total Tokens (Y) | Category |
|-----------------------------|----------------------|-------------------------|----------------|
| llama3.2_3b | 45 | 73 | Pareto-Optimal |
| llama3.2_1b | 59 | 98 | Pareto-Optimal |
| gemma3_1b | 63 | 106 | Pareto-Optimal |
| phi3.5_3.8b | 72 | 112 | Pareto-Optimal |
| phi4_mini_3.8b | 72 | 82 | Pareto-Optimal |
| phi3_14b | 73 | 86 | Pareto-Optimal |
| gemma3_4b | 74 | 101 | Pareto-Optimal |
| gemma3_12b | 79 | 101 | Pareto-Optimal |
| llama4_16x17b | 79 | 79 | Pareto-Optimal |
| devstral_24b | 80 | 75 | Pareto-Optimal |
| mixtral_8x22b | 80 | 75 | Pareto-Optimal |
| mistral-small3.2_24b | 81 | 78 | Pareto-Optimal |
| phi4_14b | 82 | 98 | Pareto-Optimal |
| llama3.3_70b | 83 | 92 | Pareto-Optimal |
| mistral-large_123b | 84 | 83 | Pareto-Optimal |
| google_gemini-2.5-flash | 86 | 78 | Pareto-Optimal |
| openai_gpt4.1 | 87 | 81 | Pareto-Optimal |
| anthropic_claude-sonnet-4.5 | 88 | 82 | Pareto-Optimal |
| gemma3_27b | 83 | 108 | Dominated |
| llama3.3_8b | 65 | 75 | Dominated |
| llama3.3_70b | 83 | 92 | Dominated |
| gemma3_12b | 79 | 101 | Dominated |
| phi3_14b | 73 | 86 | Dominated |
| phi4_mini_3.8b | 72 | 82 | Dominated |
| llama4_16x17b | 79 | 79 | Dominated |
| devstral_24b | 80 | 75 | Dominated |
| mixtral_8x22b | 80 | 75 | Dominated |
| mistral-small3.2_24b | 81 | 78 | Dominated |
| phi4_14b | 82 | 98 | Dominated |
| llama3.3_70b | 83 | 92 | Dominated |
| mistral-large_123b | 84 | 83 | Dominated |
| google_gemini-2.5-flash | 86 | 78 | Dominated |
| openai_gpt4.1 | 87 | 81 | Dominated |
| anthropic_claude-sonnet-4.5 | 88 | 82 | Dominated |

How did models do by INCOSE category?



- MCQ scores are consistently higher than OSQ scores across categories, showing that OSQ imposes a higher bar
- OSQ produces greater spread across models and categories, revealing weaknesses hidden by near-ceiling MCQ results
- Qualitatively suggests category-level qualification should not rely on MCQ alone; stronger evidence is needed where OSQ exposes larger drops

MCQ Accuracy by INCOSE Category Heatmap



OSQ Accuracy by INCOSE Category Heatmap



Limitations and Future Work



Limitations:

- **Scope:** MCQ, OSQ, hybrid evaluation, and token-efficiency analysis for SE benchmarking.
- **Excluded:** Multimodal inputs, agentic workflows, and proprietary datasets
- **Dataset:** No explicit difficulty/Bloom labels; uneven coverage across SE categories.
- **Execution:** Token limits and reasoning traces caused truncation, non-termination, or un-scorable outputs for some models.
- **Timeframe:** Results reflect the 2024 to late 2025 LLM landscape, while the framework remains extensible.

Future Work:

- **Add cognitive-depth labeling** to support difficulty-aware qualification (e.g. via Bloom's Taxonomy)
- Expand to **multi-modal artifacts** such as diagrams
- **Benchmark against human SE experts** to identify strengths, limits, and oversight needs

Bloom's Taxonomy Revised

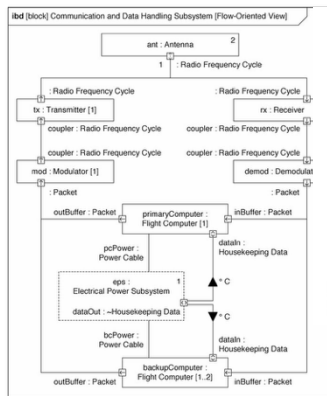
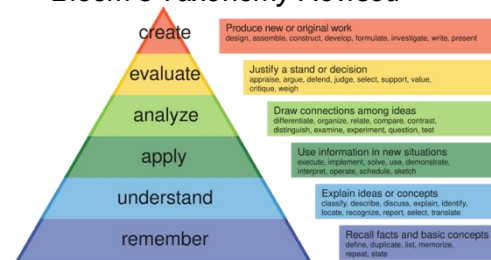
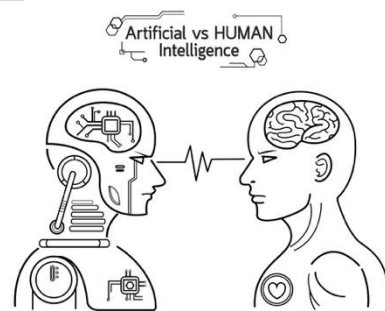
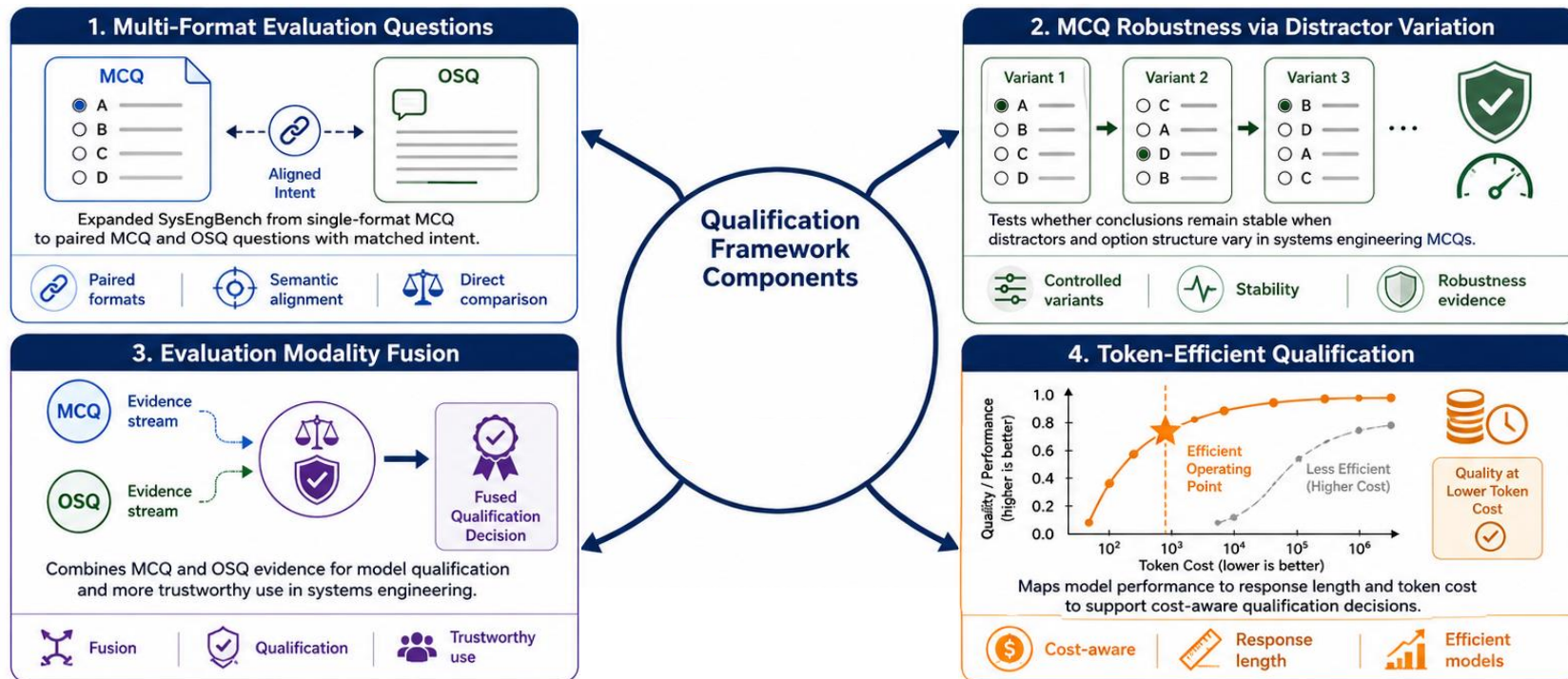


Figure 4.2 A sample internal block diagram (IBD)



Human – AI Comparison

Contributions



Outcome: Efficient task-modality guidance for systems engineering model qualification



Measure



Stress-test



Fuse evidence



Optimize cost

Conclusion



For the Systems
Engineering Community

Old question

What is the
best model?



Better question

Which models are
acceptable, for which
tasks, and at what cost?



Structured
adoption



Defensible
evidence



Task-aware
selection

For Enterprise Adoption



Faster
adoption

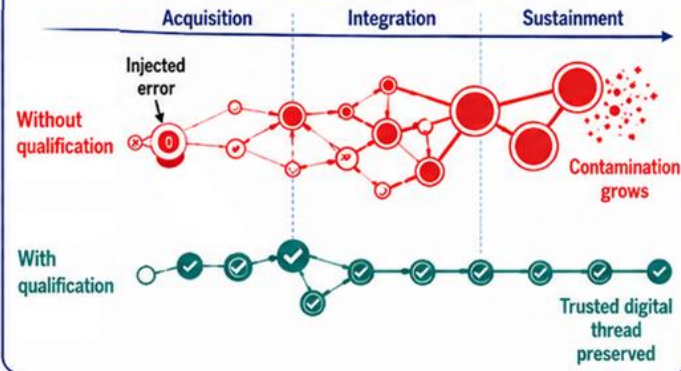


Maintained
rigor



Reduced risk of
error propagation

Error propagation across the digital thread



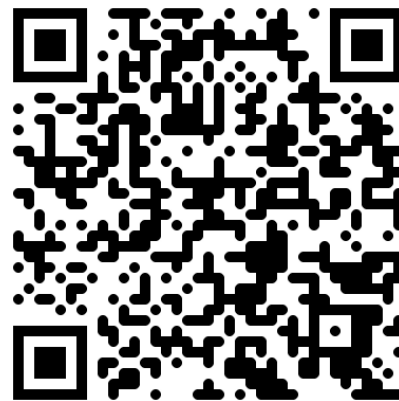
Closing Thoughts



*MCQs and OSQs should be used together for defensible LLM qualification in systems engineering; they **are complementary, not interchangeable**.*

*The next-generation systems engineer is increasingly a **hybrid SE/software practitioner**, working in text-addressable, version-controlled engineering environments.*

*Embedding LLMs into the digital thread offers **high reward**, but **also high risk** when stochastic outputs enter engineering workflows; this work supports using benchmark-qualified and governed copilots while **preserving human oversight and accountability** to reduce the risk of adoption.*



<https://ryan-a-bell.github.io/dissertation/>



Questions?

Let's Connect



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Research Topics of Interest:

- Cost Modeling
- Language Model Evaluation for Systems Engineering
- Systems Engineering in Python (*se-lib*)
- Tokenomics
- AI-Human Teaming Workflows for Systems Engineers
- AI Productivity Modeling in Systems Engineering



Backup



Current Systems Engineering language model evaluation practice over-relies on MCQ benchmarking, which can overstate readiness for systems engineering tasks that require models to construct answers and explanations in free text, leaving practitioners without a robust, cost-aware basis for model qualification.

Problem Statement